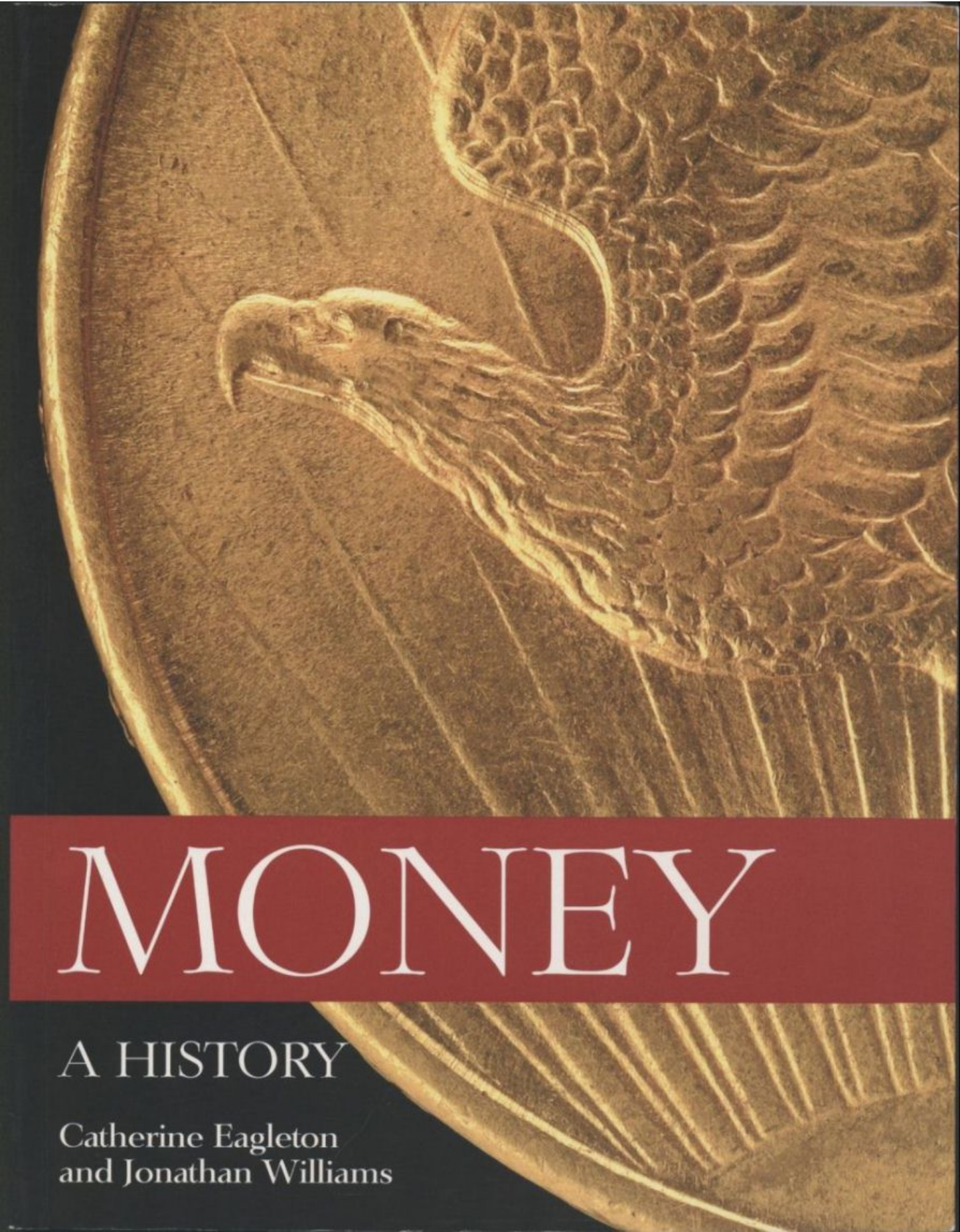


MONEY

A HISTORY

Catherine Eagleton
and Jonathan Williams

3 Medieval Europe

It gives me no pleasure to recount all of the different civil wars which afflicted the Frankish people. . . . What are you trying to do? You have everything you want! . . . In your treasuries the gold and silver are piled high. Only one thing is lacking: you cannot keep peace, and therefore you do not know the grace of God.

GREGORY OF TOURS (AD 539–94), *History of the Franks*, 5.1

The medieval world lived in the long shadow of the imperial Roman past, particularly so in western Europe where so much had changed. The end of the western Roman empire resulted in the disappearance of most public institutions that the Roman world had known, including the system of coinage and state finance which had kept money in production and in circulation. The Germanic kings who conquered the western half of the Roman Empire had little use for the complex taxation system on which the Roman state had depended for the support of its expensive infrastructure and standing army, neither of which existed in Germanic society. Precious metal continued to be of prime importance for the kings and nobles of the post-Roman world. But instead of circulating throughout the body politic as it had done in the Roman period, gold and silver began to pile up in the private treasure-houses of kings and bishops. It would be many centuries before something approaching the centralised complexity of the Roman monetary system was re-established in Europe.

Money in the wake of Rome: *c.* AD 450–*c.* 750

The barbarian peoples neighbouring the Roman Empire in the first centuries of our era seem to have regarded coins principally as bullion, but they were already familiar with money both in concept and in practice. Imperial coins are found in all the lands bordering the Rhine and Danube frontiers, and even in the far north. Among the barbarians, Roman gold and silver coins were often mounted as prestige jewellery or were melted down to make other luxury objects, but they also retained some measure of their monetary function. In times of peace there was a lively trade across the imperial frontier, and while barter served for many transactions, both gold and silver coins were used to settle imbalances and in the purchase of high-value merchandise such as furs and slaves. The leaders of the more coherent barbarian groups received imperial gifts in coin and plate, distributed with a nice regard both to the barbarians' value as erstwhile allies and to their potential threat as future enemies. The spectacular dish of 500 pounds (162 kg) of gold given in AD 451 by Aetius the

88 Silver half siliqua of Odovacar (476–93), struck in his own name at Ravenna early in his reign. Odovacar presents here the imperially correct view of his position: although proclaimed king by the barbarian troops of the 'Roman' army in Italy, he adds no royal title, wears no diadem and is shown as *magister militum* (commander of the army). (× 2)

Roman general to the king of the Visigoths may be contrasted with the modest silver bowls that ended up at second hand in the grave of a seventh-century English king at Sutton Hoo.

Large sums were also handed over to barbarian neighbours in tribute payments and in subsidies for defending the empire from more ferocious tribes further to the north and east. The scale of this drain on the imperial treasury is recorded in literary sources and attested by the many coin hoards that have been found beyond the Roman frontier. Between the years 422 and 437 the annual subsidy paid to the Huns, for instance, rose from 350 to 2,100 pounds (113 to 680 kg) of gold. But even this was not enough to prevent them from invading Italy under Attila in 452. In contrast to the earlier Celtic tribes of northern Europe who had struck coinage of their own when in a similar position, none of the Germanic barbarian kingdoms produced any official coins until after they had settled, by invitation or conquest, in the heartlands of Roman Europe in the fifth century.

In Italy the pattern was set by Odovacar, a Roman general of Gothic parent- 88
age who brought the sequence of western emperors to an end in 476. He produced a series of imperial-style denominations in gold, silver and bronze which served the needs of the monetary economy that he had inherited and successfully maintained. A multi-denominational system in all three metals was fol- 94c
lowed by his Ostrogothic successors into the sixth century. After Justinian restored Byzantine control in Italy in 552, imperial gold coins and some silver and bronze were struck again at Rome and Ravenna. Other barbarian successor states which initiated coinage in a quasi-Roman context were the Vandals in their final home in North Africa; the Lombards of Italy in their kingdom in the north and their duchy of Benevento in the south; the Visigoths first in south Gaul and then in Spain; the Franks under the Merovingian dynasty in France; and the 89
Burgundians, another Germanic people, along their eastern borders. Coinage thus continued to be produced in all the former territories of the Roman west, except for Britain. Roman coinage had already ceased to circulate there in the early fifth century, and no more coins were struck in Britain until the beginning of Anglo-Saxon minting in about 600.

89 Gold solidus of the Frankish king Theodebert I of Metz (534–48). Theodebert was the first barbarian ruler to place his own name on the solidus. Such self-aggrandising usurpation of the emperor's prerogative was noted with disapproval by the contemporary Byzantine historian Procopius. (× 4)

Nevertheless, despite the geographical continuity, the monetary systems of western Europe changed radically after the end of Roman rule. Outside Italy the coins of the barbarian kingdoms

90, 91 Gold tremissis of the Byzantine emperor Justinian I (527–65), struck at Constantinople, and a contemporary gold Visigothic tremissis in his name struck in Spain and distinguished by the cross on the chest. (× 4)

were almost exclusively gold. The reason for this concentration on high-value coins was essentially that the uses of money had altered from those in the later Roman Empire. The urban-centred economy was gradually replaced by a rural one based on the large country estate. The old Roman taxation system was less effective or had broken down, international trade continued but at a reduced level and the circulation of coin became more sluggish. In these circumstances there was less need for smaller denominations, but a continuing requirement for gold coin suitable for settling legal compensation payments characteristic of Germanic society, giving largesse, paying armies and supplying the needs of the larger-scale operator.

At first, the barbarian rulers regarded their territories as still technically within the empire and struck their gold coins in the names of the successive contemporary emperors at Constantinople. 90-91 Usually referred to as 'pseudo-imperial', these coins are often distinguished from their imperial prototypes only by their style, which is not always inferior to that of the official issues, just individually and consistently different. Lower denominations, if any, in silver and bronze increasingly bore the monograms or names of the barbarian rulers themselves. The imperial connection gradually weakened until all coins were struck in the names of the local issuing authorities.

The later sixth and seventh centuries brought change. In the less centralised Frankish kingdoms, there was a proliferation of mints and early signs of a relaxation in the Roman tradition of the ruler's monopoly of coinage in the assumption by others, particularly ecclesiastical authorities, of the right to coin. At this time, too, the wastage of gold coins was increased by the consistently unfavourable balance of trade in luxuries with the East and by the tendency of the wealthy and the Church to hoard what they had. Dwindling stocks of gold could not be augmented with new supplies as there were no significant sources of raw metal in the West, and the former regular infusions of imperial *solidi* in the form of subsidies from the Byzantine Empire ceased. There was a serious shortage of bullion and, throughout the barbarian kingdoms, gold was increas-

92 Gold tremissis struck in Merovingian France by the moneyer Lobosindus, *c.* 605–15, with the formula *Racio Dom[in]i* (treasury of the king). (× 2)

93 Silver penny of Archbishop Ecgberht of York (732/4–66) with King Eadberht of Northumbria (737–58), showing a full-length figure of the archbishop. The Church was involved in coin production from its beginning in England. (× 2)

ingly confined to the smaller denomination, the *tremissis* (worth one third of a solidus). The tendency, evident from the beginning, for the weight and fineness of their gold coins to fall away from the old imperial standard was considerably accelerated. The pace and extent of this process varied with the individual circumstances of each successor kingdom.

The problem became more acute as the demand for coin began to rise. Successful barbarian dynasties, despite internecine warfare, created more settled conditions and developed fiscal systems which promoted commercial life and the use of money. In addition, the newly granted markets and fairs called for a coin less valuable than the gold tremissis. At this point, the growing influence of the Christian Church was also important, on the one hand increasing the drain on bullion by converting gold coins into plate and ornaments, while on the other stimulating the circulation of money in its roles as collector and dispenser of alms, patron of craftsmen and as a participant in the revival of trade.

These developments are particularly clear in Merovingian France where, 92
during the first half of the seventh century, the tremisses and the rare regal solidi became rapidly paler as the gold was adulterated with silver. With a brief overlap, they were replaced in about 675 by silver coins, some of which carry the inscription DENARIVS. These were of approximately the same size and weight as the tremissis. Anglo-Saxon England, dependent on continental bullion, 93
could not sustain a gold coinage after the Merovingian standard had collapsed, and changed over to silver about the same time. The source of the metal is a problem, as silver was previously scarce. Long-held stocks in the form of plate no doubt contributed, but it seems likely that newly mined silver became available, most probably from the Melle region of western France, which was to be much more productive later. These west European silver pennies of small thick fabric are unhistorically, but usefully, known as *sceattas* (an Old English word with a derivative meaning of 'treasure'). Central to the rise in commercial activity and the use of coined money from about 700 were the Frisians, based in the Netherlands, who traded in slaves as well as exchanging northern commodities for southern luxuries. Their sceattas rapidly became a major element of northern European currency. Another series of early eighth-century sceattas is attributed, still with some reserve, to the southern Danish town of Ribe on the North Sea coast, an important outpost of this trading system.

During the second quarter of the eighth century the weight and particularly the fineness of the sceattas declined. Some of the latest issues in England have such a low silver content that they appear almost like bronze coins. The reasons for this collapse of the standard are complex. Problems of bullion supply and an adverse balance of trade again played their part, but perhaps most important was the suppression of the independent pagan Frisian kingdom by the Christian Carolingians.

The uses of money

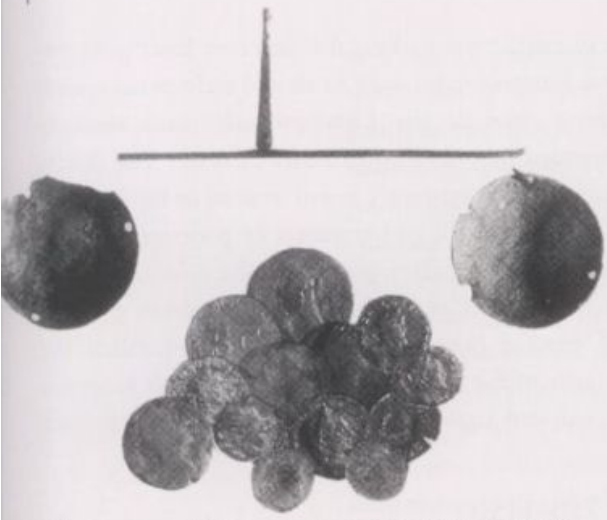
The social institutions of the Germanic peoples who settled in the former western Roman Empire, particularly compensation payments for various crimes and aristocratic gift-giving, were facilitated by the adoption of money and stimulated its use. Payments to troops, purchases of luxuries and property and donations to the Church all employed coined money. While barter continued, money was also used in commerce. As systems of taxation, tolls and customs were developed in the established successor states, money - while not a necessity - was increasingly used. It was high denominations, essentially gold, that were needed for these transactions. Even silver coins, where they existed, were relatively high-value and it was only in North Africa and Italy that lower-value copper coins continued to be struck to facilitate small-scale market purchases in an urban economy.

94a Gold and garnet Anglo-Saxon pectoral cross set with a gold solidus of Heraclius and Heraclius Constantine of the period 613–32, found at Wilton, Norfolk. This pendant adapts the pagan tradition of using coins for jewellery to the devotional purposes of a recent Christian convert. (× 1½)

94b Leaf from the ivory diptych of Orestes, Roman consul, AD 530. The consul sits between Rome and Constantinople personified. Above are medallions of the Ostrogothic rulers Athalaric (left) and his mother Amalasuntha (right). Below, servants pour money from sacks, symbolising the consul's customary largesse to the populace and demonstrating the continuity of Roman tradition under the Germanic kings (London, Victoria and Albert Museum).

94c Copper follis (40 nummi) of Theodahad, king of the Ostrogoths (534–6), struck in Rome. The king is shown wearing a jewelled *spangenhelm*, the helmet-like 'crown' of Germanic kings. (× 1½)

94d Silver quarter siliqua of Pope Gregory III (731–41) with the Byzantine emperor Leo III, struck at Rome. The pope's monogram appears on the reverse. The coinage reflects the passing of real authority in Rome from the Byzantine emperor, still in nominal control, to the popes.



94e Bronze pan-balance and Roman coins used as weights, found in a pagan Anglo-Saxon warrior's grave at Dover, sixth century AD. Anglo-Saxon coins were not yet being produced and such equipment was used to weigh small amounts of gold and foreign gold coins. (× ⅔)

94g Hoard of 100 Frankish and Anglo-Saxon gold tremisses, buried (with a few ornaments) *c.* 645 at Crondall, Hampshire (from electrotypes of the originals now in the Ashmolean Museum, Oxford). The round sum of 100 gold shillings is thought to represent a *wergeld* (compensation payment for a killing). (× 1¼)

94f Detail of a drawing from the Eadwine Psalter, a mid-twelfth-century English copy of a Utrecht original of *c.* 820, illustrating Psalm 123: 'Behold as the eyes of servants look unto the hand of their masters . . . .' The king watches two servants tip money from sacks onto a table while another weighs the amount due to each waiting soldier (Cambridge, Trinity College).

94h Hoard of 118 Anglo-Saxon and continental silver pennies (sceattas), buried *c.* 730 and found at Woodham Walter, Essex. The hoard, found close to a road leading inland from the port of Maldon, contained many Frisian coins from the area of the Rhine estuary, suggesting that this was money used in trade.

95 Base-silver denier of Nemfidius, Patrician of Provence (*c.* 700–10), struck at Marseilles. The growing independence of the administrators of Provence at the expense of the weak Merovingian kings led them to replace the royal name with their own on the coinage. (× 1½)

Meanwhile, the Visigoths in Spain were making do with ever baser gold tremisses and no silver until their issues were brought to an end early in the eighth century by the Arab conquest. In Italy the Lombard gold coins similarly declined in fineness, supplemented by only limited issues of silver. The duchy of Benevento in southern Italy, which apparently began striking in the late seventh century, sustained a coinage of solidi and tremisses of poor-quality gold into the ninth century, latterly alongside silver pennies of the reformed broad type inspired by those instituted by the Carolingian ruler Pepin about 755.

The monetary history of western Europe in these centuries parallels the transition from the essential unity of the empire to the plurality of the successor barbarian regimes, all with a shared Roman inheritance but with their own distinctive identities.

The age of the penny: *c.* 750–1150

At the dawn of the Carolingian age European currency was in transition: a vestigial Roman monetary system of coinage and taxation struggled on in the south, while in the north-west the framework of its medieval successor was already in place. The trend towards decentralisation characteristic of the new order was, however, about to be reversed, if only temporarily. The precursor of this development was the fall of Spain to the Arabs in 711–15. The whole of the peninsula became part of the Islamic world, adopting the monetary system used in the Umayyad caliph's other territories, which stretched from the Indus to the Atlantic (see fig. 134c). The eighty or so mints of the preceding Visigothic period were reduced to a single mint at Cordoba which served the entire country. There was a clear contrast between the monetary institutions of the post-Roman barbarian West and the Islamic world.

96 Silver denier of Emperor Louis the Pious (814-40), struck at Melle, Aquitaine, 819-22. The mint name *Metallum* on the reverse means 'mine' in Latin, and refers to the rich local source of silver. (× 1½)

The signal for change in France was the overthrow of the last effete Merovingian rulers by the vigorous founders of the Carolingian dynasty. Its first king, Pepin the Short, carried out a radical reform of the coinage in about 755, by which the higher weight and silver content of the original penny were restored. The new money, also distinguished from the old by a broader, thinner fabric, was to remain the standard silver denomination in Europe for the rest of the Middle Ages. The reform also tightened royal control over minting operations, symbolised by the presence of the ruler's initial or name and title on all coins. The number of mints was reduced and coins of uniform design were soon struck throughout the king's or emperor's dominions, including northern Spain recovered from the Arabs and in the former Lombard kingdom in north Italy. The volume of Carolingian coinage was high, sustained by fiscal and commercial success and, in particular, by the effective exploitation of the Melle mines 96 in western France.

The political situation in southern England also favoured increased centralisa-

tion and an explicitly regal currency. The largely anonymous sceattas were superseded by broader pennies of Carolingian inspiration and of restored weight and fineness, struck in the name of Offa, king of Mercia (757–96), who had made himself master of all England south of the Humber. Fewer broad pennies were issued than sceattas in their heyday, and the total amount of silver in circulation was smaller, even after the greater intrinsic value of the new coins is taken into account. This decline, continuing a trend which had begun well before the reform, is probably to be explained by an unfavourable balance of trade and the absence of significant sources of raw silver to make up the deficit. A lack of bullion was even more evident in the local money of northern England, which consisted of coins with an ever decreasing silver content but which nonetheless met the needs of a lively internal economy. When fine-metal money was required, southern English and Carolingian coins were used, but the conquest of the Northumbrian kingdom by the Scandinavian invaders in 867 extinguished minting there for a generation.

The Viking menace was a factor of growing importance in ninth-century 97
Europe. It caused, at the same time, both the expansion and the weakening of west European money supply. Many new Carolingian mints were opened to meet the need for coin to pay the massive tributes demanded by the raiders. In southern England, once more politically divided, the volume of coin produced also rose steeply. More fundamentally, Viking activity disrupted agricultural and urban life and also the internal and external trade on which earlier prosperity had been based. Although severely strained, the Carolingian system, with its access to greater reserves and new bullion, coped better than the English one, which came near to collapse with pennies containing only 25 per cent silver or less.

During the same century the Vikings were developing their own trade routes across Russia to Constantinople and the Islamic caliphate further to the east. This commerce was heavily dependent upon the sale of slaves for silver in the form of Arabic *dirhams*, which were brought back to the north in large 97a
numbers. The establishment of settled Viking communities in the British Isles in the later ninth century saw the extension of their trading network westwards across the North and Irish Seas, and their rulers soon produced money for the same range of purposes as their longer-established neighbours and former enemies.

Although there had been fundamental internal changes to the monetary system in large areas of western Europe, the geographical extent of regular coin use had scarcely changed since the days of the later Roman Empire. There had been an early extension of minting into Frisia, some advance into Germany following the Carolingian conquests and occasional issues in Scandinavia. But a rapid expansion eastwards occurred in the early tenth century when power in

The Vikings

The traditional view of the Vikings' role in monetary history from the eighth to the eleventh centuries is of plunderers and extortionists satisfying their thirst for silver by ruthlessly preying on the wealth of more financially advanced societies. Recent scholarship has been concerned to highlight their more positive activities in this area, particularly their contribution to the development of towns and trade. It is difficult to strike a just balance between these aspects: both represent part of the truth and are interrelated. In western Europe the raising of gelds (taxes) to pay off the Vikings stimulated the development of a money economy among their victims, and the exacting of tribute did not preclude commercial relations. The Vikings were responsible for developing trade routes through the eastern Baltic with Kiev, Russia and Central Asia, but a major source of their profit was the sale of slaves seized in raids.

Once established in settled communities in their conquered lands, the Vikings adopted existing monetary systems, or created them from scratch, as at Dublin, often using native or imported experts to operate them. The initial impetus may have been prestige or the need to make official gifts, but coinages for normal fiscal and commercial purposes usually followed. In their homelands the Vikings were mainly content to use foreign money and were slow to introduce exclusive national coinages because it took time to develop the institutions needed to run them successfully. But settled economies did not tame the Vikings' military energy or end the urge for expansion, and their heirs continued to fight for land and fortune under their more respectable name of Normans.

97a Silver dirham of Caliph Al-Mu'tamid (AH 256–79/AD 870–92), struck at Arminiyah, AH 277 (AD 890–91), from the Cuerdale (Lancashire) hoard buried *c.* 905 (see frontispiece). The Islamic coins found in England had travelled from the caliphate by way of Viking trade routes along the Russian river systems and through Scandinavia. (× 1½)

97c Silver penny of Anlaf Guthfrithsson (939–41), struck at York, showing the pagan motif of a raven. The Vikings, who intermittently controlled York during the tenth century, participated fully in the trading activities of the city and their coins were often struck by moneyers who had worked for their Anglo-Saxon predecessors. (× 3)

97b Srebrennik (silver coin) of Jaroslav the Wise, Prince of Kiev (1015–54). People of Scandinavian descent had settled at Kiev on the Dnieper, a focal point of Viking trade with Constantinople and the east, in the ninth century. Denominations and designs were Byzantine in inspiration (Stockholm, Kungl. Myntkabinettet, Statens Museum för Mynt Medali och Penninghistoria). (× 1½)

97d Silver penny of Sihtric III Silkbeard, Norse king of Dublin (993/4–1042), struck by the moneyer Fastolf *c.* 997. This issue, based on Aethelred II's crux-type penny, marked the start of minting in Ireland. Viking Dublin was an important centre for trade. (× 1½)

97e Main Viking trade routes.

97f Silver penny of Olaf Skötkonung of Sweden (*c.* 994–1022), struck at Sigtuna, *c.* 995. The earliest coins naming the kings of Sweden, Denmark and Norway were made about the same time. They all copy the crux-type penny of Aethelred II of England, used to pay large amounts of money to the Scandinavian invaders. (× 1½)

97g Silver penny of Cnut, Danish king of England (1016–35), struck at London. In 1018 Cnut exacted a geld to pay off his invasion fleet. In this way large numbers of English coins reached Scandinavia, where, like this example, they were often 'pecked', i.e. tested with a knife-point. (× 1½)

97h Silver pfennig of Otto III (as king in Germany, 983–96, with his grandmother Adelheid acting as regent), struck at Goslar after 991 in metal from the Rammelsberg mines. This coin is pecked and had circulated in Scandinavia. Many German coins travelled northwards in trade, but tribute and plunder were also involved. (× 1½)

97i This stone at Yttergärde in Uppland, Sweden, was dedicated in the 1020s to the memory of Ulv of Borresta. The runic inscription says that Ulv 'has in England taken three gelds. The first which Tosti paid. Then Thorkell paid. Then Cnut paid.'

Germany passed to the Saxon dynasty with its centre in Bavaria. This political change was complemented by the discovery, around the middle of the century, of rich silver mines in the Harz mountains near Goslar in Saxony. The spread of money use and minting continued as peoples on the eastern and northern fringes of the German empire in Bohemia, Hungary, Poland and Scandinavia 98 developed settled states which first used foreign coin and then saw the advantages of converting this into money of their own. The adoption of Christianity

98 Silver penny of Boleslav II, Duke of Bohemia (967–99), struck at Prague. The obverse design, showing the Hand of God, was copied from a coin of Aethelred II of England of the early 980s. Anglo-Saxon coins reached Bohemia in trade, but the source for this type may also reflect political and religious affiliations, as Boleslav's wife, Emma, may have been English. (× 5)

in these lands also helped to promote the use and production of money. By the year 1000, coins were being struck from Dublin to Kiev, although Scotland did 97 not have its own coined money until more than a century later. Many of these early issues were small and intermittent, often produced for distribution rather than circulation; normal monetary needs were met by foreign coin which arrived in tribute or trade. National coinages developed more fully in eastern Europe from the eleventh century but often on a relatively limited scale compared with those of the more mature, and mine-owning, states in the West.

From the tenth to the early twelfth centuries the decentralising trend in France and Germany was intensified. First the French king and then the German emperor granted coining rights to ecclesiastical and lay magnates whose wealth and power had increased at the expense of central control. The strongest

of these feudal coinages, such as those of the Abbey of Tours, the Counts of Anjou and Champagne, or the Archbishops of Cologne, dominated the money 99–100 in regions far beyond the borders of their own territories. The need to maintain confidence and acceptability caused many coinages to become 'immobilised', that is, to continue to use the same basic designs and inscriptions for many years.

In the earlier part of this period the use of money, the volume of currency and the velocity of circulation all increased. Although revived Viking activity played a part, this expansion was principally based on renewable silver stocks and the growth in local and international trade exemplified by important fairs such as those in Champagne. By the mid-eleventh century, however, the Goslar mines in Saxony had been worked out and the resulting shortage of new silver brought a reduction in minting and money supply throughout Europe. There was also a related tendency for issues to decline in both weight and fineness. Although the volume of production had also decreased in England, the standard 101 of the coinage was maintained, partly as a result of a favourable trade balance probably already involving the export of wool. Also contrary to the trend elsewhere in Europe, a steady reduction in the number of mints in England began in the later eleventh century. Neither did feudal currencies develop as they did in France, and the English royal monopoly established by the West Saxon dynasty survived both the Danish and Norman conquests. A few ecclesiastical authorities and lay magnates retained minting rights, but their coins conformed in all respects to regal issues. Only during the civil war in the 1140s were baronial coinages struck in England, but they did not survive the restoration of peace.

The sole currency coin throughout most of northern Europe during this period remained the silver penny. As market transactions became more common there was a growing demand for smaller denominations. Round halfpennies were struck from the mid-ninth century, but only intermittently, and needs were otherwise met by cutting the penny into halves and quarters – and even smaller parts in eastern Europe. Gold coins were struck almost exclusively for ceremonial use, although Byzantine and Arabic gold coins did occasionally reach north-west Europe to provide the bullion for these local issues.

In north and central Italy money developed largely along north European lines with the silver penny as the standard currency unit. Coinage remained the monopoly of the Carolingian rulers

99 Base-silver denier of Thibaut II, Count of Champagne (1125–52), struck at Provins. This coinage owed its success to the income generated by Champagne's international fairs. (× 1½)

100 Silver pfennig of Sigwin, Archbishop of Cologne (1079–89). The coins of Cologne were first regal, then struck in the joint names of the German emperor and the archbishop, and later, as here, in that of the archbishop alone, illustrating the extension of coinage rights. (× 1½)

101 Silver penny of William I 'the Conqueror' (1066–87), struck at Gloucester by the moneyer Ordric. The reverse illustrates the control measures, inherited from the Anglo-Saxons, used to maintain the high quality of the coinage. The inscription named the moneyer and the mint so that the issuer of coins deficient in weight or fineness could be traced and punished. (× 1½)

102 Gold dinar of Offa of Mercia (757–96), copying a gold dinar of the Abbasid caliph Al-Mansur struck at Baghdad in AH 157 (AD 773/4). Offa's coin may have been struck to fulfil the promise he made in 786 to send an annual gift to the pope, but its condition shows that it was used in commerce. (× 2)

and then of their German imperial successors. Minting rights were not granted to other authorities until later in the twelfth century, beginning with the city of Genoa in 1138. The Papacy issued silver pennies in the joint names of current pope and Carolingian emperor from the eighth to the later tenth century, but never on a large scale. In south Italy and Sicily the currency remained outside the general European system. It reflected the region's position as a point of contact between the Islamic and Byzantine empires in including both gold and copper coins whose types were often derived from Arab or Byzantine models. The Normans who carved out principalities for themselves there in the eleventh century continued to issue gold and copper coins in the local tradition, but also incorporated a base-silver coinage into their system.

During the tenth and eleventh centuries the Arabs' power in Spain had declined as they first parted from the central caliphate and then broke up into many petty states, some of which issued coinage, generally on a very limited scale. In the late eleventh century they joined the empire of the Almoravid rulers of North Africa, which revived the issue of gold and silver coins. Northern Spain had been lost to Christian states, from which emerged the kingdoms of Leon, Aragon and Navarre, alongside the former Carolingian territory of Barcelona, now independent under a powerful dynasty of counts. These states produced silver pennies on the French model, but some also struck gold coins based on the issues of their Arab neighbours which they had obtained through normal trade and in tribute payments.

By the middle of the twelfth century the period of monetary recession in western Europe was about to be replaced by one of great expansion as a result of an infusion of new silver from the massive deposits discovered at Freiberg in Meissen. But before continuing with this story, we should take a brief look at the long and important history of the Roman Empire in the east, centred on the city of Constantinople, which has come to be known to historians as the Byzantine Empire.

Byzantium

[It] is accepted everywhere from the ends of the earth. It is admired by all men and in all kingdoms, because no kingdom has a currency that can be compared to it.

COSMAS INDICOPLEUSTES (6th century AD)

For centuries the Byzantine Empire lay between the worlds of medieval Europe and Islam, a crucial zone of influence and exchange as well as a great civilisation in its own right. Its coinage provided the model for the first independent currencies of its neighbours and successors, while for over half a millennium the gold solidus or *nomisma*, the bedrock of its monetary system, ruled supreme as

the principal trade coin of the Mediterranean world, the 'dollar of the Middle Ages' as it has been called. It was also familiar as the *bezant* over a much wider area across Europe and Asia. When Western kings wished to make religious offerings in gold coin, they used bezants.

103 Gold nomisma (solidus) of the Byzantine emperor Constantine VII and his son Romanus II (co-rulers 945–59), mint of Constantinople, showing the image of Christ. From the mid-ninth century religious images increasingly dominated the coinage. The figure of Christ became the normal obverse design, relegating the emperor to the reverse. (× 1½)

When 'Byzantium' begins is a matter of continuing debate. For the coinage a range of developments in the sixth century provides a useful marker. In about 498 the emperor Anastasius introduced a series of large copper coins to replace the tiny late Roman bronzes. Christian emblems such as angels and crosses replaced the pagan figure of Victory on the backs of the gold coins; a full-facing bust of the emperor became the normal imperial image on the front, while ornate robes replaced military costume as the usual imperial attire. In a long transition during the seventh century Greek took the place of Latin in coin legends, the most obvious consequence of which was the replacement of 'Augustus' by 'Basileus' as the imperial title.

A by-word for stability, the Byzantine gold coinage remained, from its origin 103
as the solidus of Constantine the Great, unchanged in standards for centuries and seemingly always in plentiful supply. Its position was shaken by heavy debasement in the mid-eleventh century, but a revival of standards in 1092 under Alexius I Comnenus restored its reputation until the thirteenth century.

This coinage, like that of Rome before it, probably owed its form to the needs of the state rather than the interests of commerce. The constant availability of a stable gold coinage enabled the state to function, accumulating cash through taxation to pay for its standing army, civil servants, buildings, ceremonials and foreign subsidies – the armies alone may have regularly absorbed over half the state's income. The Byzantine state taxed and spent on an enormous scale in comparison to contemporary western Europe. A land tax was the basis of the system, and to pay it landowners sold their surplus produce – often to the state itself to supply the army, the administration and the imperial post.

Taxation in gold coin, rather than in kind, simplified collection and distribution, and also facilitated the building up of reserves of treasure. Treasure was important because the Byzantine state was not able to have much recourse to credit and loans – in a crisis the emperor had to look to his regalia, the Church or great magnates to tide him over.

While the gold coinage existed to fulfil the purposes of the state, it inevitably played a role beyond that in daily life. Furthermore, supporting it were lesser denominations, mostly in copper, which permitted smaller-scale monetary transactions to occur. In any case, provision of such a coinage helped maximise the state's holdings of treasure: taxes had to be paid in gold, but the state could make payments (including change out of gold offered for tax) in copper.

Initially there was a range of smaller gold and copper coins, though from the mid-eighth century the only lesser coin normally available was the copper *follis*.

Intermediate silver coinages were only periodically provided, though the *milaresion* (1/12 nomisma) was in reasonable supply in the ninth and tenth centuries. In general the empire remained predominantly agricultural and rural, and, outside Constantinople and a few other centres, trade would have been confined to agricultural produce or low-level dealings and was probably highly seasonal. Merchants were of low status and subject to state regulation. The relative lack of low-level coinage between the seventh and twelfth centuries suggests a restricted urban economy.

PRINCIPAL DENOMINATIONS OF THE BYZANTINE EMPIRE

FIFTH TO EIGHTH CENTURIES		
Gold	*Silver*	*Copper*
solidus/nomisma	hexagram	follis (40 nummi)
semissis (1/2 nomisma)		half-follis (20 nummi)
tremissis (1/3 nomisma)		decanummium (10 nummi)
		pentanummium (5 nummi)

EIGHTH TO TWELFTH CENTURIES		
Gold	*Silver*	*Copper*
nomisma	milaresion (1/12 nomisma)	follis (1/288 nomisma)

TWELFTH TO THIRTEENTH CENTURIES			
Gold	*Electrum*	*Base silver*	*Copper*
hyperperon	aspron trachy (1/3 hyperperon)	trachy (1/48 hyperperon, later 1/120)	tetarteron (uncertain value)
			half tetarteron (uncertain value)

104 Gold hyperperon of Emperor Alexius I Comnenus (1081–1118), struck after the reform of 1092. In the eleventh century the old nomisma denomination became broader and distinctly concave in shape. In the mid-century the coinage suffered serious debasement, until 1092 when Alexius I introduced a new gold coin, the hyperperon. This had the concave form of the debased nomisma, but its fineness was restored to a high level (20½ carats). (× 3)

In 1092 Alexius I's reform supported the newly restored gold coinage (the *hyperperon*) with a wider range of 104 pieces in electrum, base silver and copper, somewhat broadening the character of available money. In the last years of the diminished empire, from the recapture of Constantinople in 1261 to the final Ottoman capture of Constantinople in 1453, the once mighty bezant disappeared for good, replaced by a poor coinage in silver and copper.

The later Middle Ages in western Europe: c. 1150–1450

Men were subtle enough to devise the use of money to be the instrument of natural riches which of themselves administer to human need. . . . For money does not directly relieve the necessities of life, but is an instrument artificially invented for the easier exchange of natural riches. And it is clear without further proof that coin is very useful to the civil community, and convenient, or rather necessary, to the business of the state.

NICHOLAS ORESME (*c.* 1320–82), *A Treatise on the Origin, Nature, Law and Alterations of Money*

105 Avarice, from a medieval manuscript illumination. Images of money, particularly its hoarding and testing, have featured in Christian art with largely negative connotations. Above all, the miser is the classic symbol of avarice, one of the Seven Deadly Sins, as depicted in this manuscript: demons hover around him, both inspiring and reflecting the spiritually destructive allure of gold (British Library).

By the twelfth century the issue of coined money was an established factor throughout most of Europe, and minting was becoming more centralised. Lands coming late to coinage, such as the Scandinavian and eastern European kingdoms, retained a relatively centralised approach to the provision of money along with virtual royal monopolies; the same held good for the Christian kingdoms of Spain as they expanded at the expense of the Arabs. In England, where royal control of coinage had always been retained, the number of issuing mints was gradually reduced to a handful, dominated by London. In France the previously insignificant regal coinage expanded its role greatly in national currency at the expense of the coins of the feudal lords, which were restricted to the lands of their issuers. In the lands of the Holy Roman Empire (Italy, Germany and the Netherlands) the number of coin issuers continued to expand, but many of the most significant and influential were already established.

This interest in the control of minting in the twelfth century reflected the increased role that money was playing in European society. At a time of rising population, town and city growth and expanding commerce, both local and international, coinage had to be adapted to fulfil increasing monetary needs. New silver supplies both augmented the overall money supply and enabled a relatively high-value, fine silver coinage to be restored to areas such as north Italy and the Netherlands, where economic activity was making its greatest advances.

The spread of the new silver coin varied in the context of local coinage, but there was a strong tendency at first to make it worth twelve of the local pennies (or a suitable fraction or multiple thereof), thus fitting it into the widespread system of account of 12 denarii (pennies) to the solidus (*soldo*, *sou*, shilling), and 20 solidi to the *librum* (*livre* or pound). This system formed a money of account and played a vital part in late medieval money beyond its convenience in reckoning, acting as a common denominator for the changing values of coins as the monetary systems of Europe became more varied and complicated.

106 Silver denier tournois of Philip II Augustus, King of France (1180–1223), showing a stylised castle. The abbey of St Martin of Tours was an important coin issuer in western France. Philip Augustus took over the issue of its popular coin from 1204 and it joined the denier parisis as official royal coin. It also became the basis of the national money of account. (× 1½)

In the thirteenth and early fourteenth centuries larger silver coins, multiples of the old penny denomination, spread into most European coinage systems, usually under a name derived from the Italian term *grosso denaro*, or *grosso* ('big penny'), which in English became the groat. The pioneering Venetian grosso 107
was introduced in around 1202 and equalled 24 local denarii, though both it and the Florentine grosso of 12 denarii in fact weighed little more than a contemporary English penny. The French *gros tournois* (introduced in 1266) was worth 111
12 *deniers tournois* (one sou) but weighed twice as much as the grosso, while the 106
English groat, of similar size to the gros tournois, was worth four pence and only found its niche in English currency from 1351.

The silver that had allowed this new pattern of coinage to develop came from 108
a series of new mines discovered during the twelfth and thirteenth centuries. The first was at Freiberg in Saxony in the 1160s, to be followed by others in 109
Italy, Bohemia and the Alps. The last new mine, at Kutna Hora in Bohemia, discovered in 1294, would produce 20–25 tons (20,300–25,400 kg) of silver annually in the early fourteenth century.

107 Silver grosso of Venice, issued by the Doge Giacomo Tiepolo (1229–49), who is represented receiving a banner from St Mark, patron of Venice. In the late twelfth century northern Italy was well placed to attract new silver supplies, and was also commercially advanced and urbanised. However, its coinage consisted solely of tiny base-silver denari. The pioneering Venetian grosso, probably introduced in 1202, was a fine silver coin, bigger than any seen in Europe since antiquity, and worth 24 of the local denari. (× 1½)

The impact of new silver on Europe's money supplies was enormous. For instance, it has been estimated that the amount of coined money in circulation in England in 1300 was not equalled again until about 1600. Nevertheless, despite the new silver coins, high-value dealings were still often facilitated by the use of silver ingots. These may sometimes have been stamped to indicate their adherence to silver of the coinage standard of, say, the Venetian grosso or English sterling penny. Gold dust in sealed bags, weighed by the ounce, may also have been thus employed in the lands bordering the Mediterranean. Much of the dealing of money-changers would have involved exchanges between local coin and marks of silver or ounces of gold. Only gradually did the revival of gold coin displace this traffic.

108 Silver pfennig of Philipp I von Heinsberg, archbishop of Cologne (1167–91). Minting at Cologne expanded from the 1170s, from virtually nothing to over 2 million pfennigs a year, following the opening of the Freiberg silver mines. The Cologne pfennig came to be one of Germany's dominant coins, used widely in the Rhineland and beyond throughout the thirteenth century. (× 1½)

109 European mines in the tenth to sixteenth centuries.

The lowest level of transactions probably also occurred without much use of coinage. Barter and the offsetting of small-scale loans were presumably commonplace in local communities, and would long remain so. The provision of small change was, however, affected by the new types of coin. Increases in the money supply made the use of coin more familiar and regular. Complaints about the shortage of small denominations became common in the later Middle Ages, as demand conflicted with the reluctance of mints to give due attention to this part of the currency, which was more labour-intensive and less profitable to produce. These complaints may in part have been encouraged by increased expectations of the roles money was to fulfil. It is possible that people resorted to unofficial tokens in base metal to supplement the lack of low-value official coinage, though this has not been adequately demonstrated for the medieval period.

The old penny equivalents were, however, gradually debased in their silver content, and in many areas their value steadily declined against the fine silver money. As the proportion of copper to silver in their make-up increased, they earned the name 'black money'. This could have a knock-on effect on

110 Silver pfennig of the archbishops of Salzburg, Friesach mint, *c.*1216(?)–87, showing an angel under a canopy. Heavy production at Friesach began around 1195, reflecting the output of its silver mines. Until the 1230s Friesach and its neighbouring mints were immensely productive. In the thirteenth century Friesacher pfennigs dominated currency from the eastern Alps to the Carpathians. (× 1½)

111 Silver gros tournois of Louis IX, King of France (1226–70). This was the first large silver multiple north of the Alps, worth 12 deniers tournois. It was successful and became a familiar coin internationally, particularly in the Rhineland and the Netherlands. (× 1½)

112 Silver penny of Edward I, King of England (1272–1307). England's export trade ensured that the country had ready access to bullion, and its currency expanded to match the general growth in silver supplies from the late twelfth century. The type introduced by Edward I was to be long-lived and internationally popular. (× 1½)

113 Silver sterling of Jean d'Avesnes, Count of Hainaut (1280–1304), showing him wearing a garland of roses. English sterling pennies became plentiful in the Netherlands, and to defend or expand their mint profits local princes began to produce their own versions. In the 1290s perhaps 90 million continental sterlings were issued, in contrast to the 9 million English ones struck. (× 1½)

prices, as these coins were often the base coin of the money of account, but this process also encouraged the use of coin for ever smaller transactions. Where the base coin's value became too small to be functional it disappeared altogether and a multiple usually took its place in the system. Those lands, like England, which maintained fine silver for their smallest denominations (the halfpenny and farthing) saw these coins shrink to an impractical size while still retaining significant purchasing power.

114 Gold augustale of the emperor Frederick II, as king of Sicily (1197–1250). Sicily had been part of the Islamic world, with a coinage of small gold tari which was continued by the Normans. In 1231 Frederick II added the augustale, a coin of 20½ carats gold, on the standard of the Byzantine hyperperon and of the African gold from which it was made. It became familiar in Italy, paving the way for the revival of gold coinages in the west. (× 3)

The new silver multiples proved only a partial solution in the provision of high-value money. The next step was the revival of gold coinage and the creation of a multi-metallic system (gold, silver and often base silver) covering a range of denominations. A usable gold coinage of course required sufficient quantities of gold. Europe's own supplies were limited, with the most important sources being attached to the kingdom of Hungary, in particular Kremnica (modern Slovakia), which became productive in the 1330s. More importantly, in the thirteenth century, Italian trading cities, particularly Genoa, Florence and Pisa, increasingly captured the gold trade with North Africa, the gold coming ultimately from either the western Sudan or the west coast of Africa and being transferred across the Sahara to the Islamic lands of North Africa.

In 1251–2 both Genoa and Florence launched gold coins, the *genovino* and the *florin*. The latter was to become enormously important in international trade for the next century, spreading the concept of gold coinage across Europe and inspiring a host of imitations. Florins of Aragon were important in the western Mediterranean, florins of Lübeck in the Hanseatic ports of the Baltic, and florin-based *gulden* provided a common currency for the varied principalities of the Rhineland. The florin's closest rival was to be the Venetian *ducat*, introduced in 1284, which gained the dominant role in eastern Mediterranean trade. As familiarity with gold coin spread, Western rulers in the fourteenth century adopted the new concept, though it often took much trial and error for them to produce coins that reflected the relative values of gold and silver sufficiently well to permit both to remain effectively in currency in a fixed relationship.

114–5

115 Gold florin of the city of Florence, 1252–1307, with a lily as its city badge. By 1252 enough African gold was passing through Florence for it to launch its own gold coinage. The florin was pure gold, initially valued at 1 lira in Florentine reckoning. Its international importance was immediate, both in the Near East and western Europe.

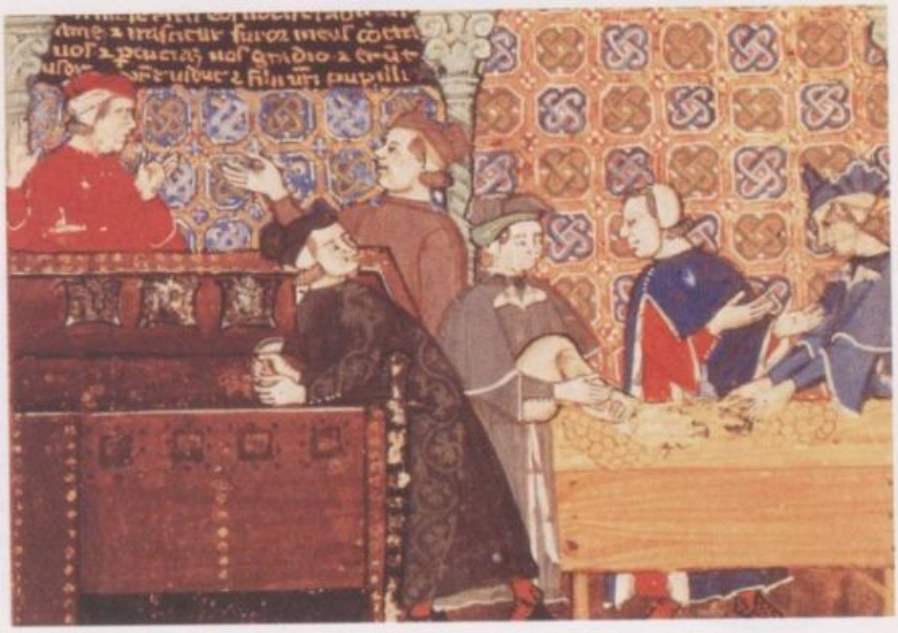
116 Gold ducat of Giovanni Dandolo, Doge of Venice (1279–89), showing him kneeling to receive a banner from St Mark. Venice had less immediate access to the North African gold trade, and may also have continued its use of Byzantine coin. Only in 1284 did it produce its own, similar in standard to the florin and apparently primarily for international use. (× 2)

In the wake of Kutna Hora, the last great new bullion source of the central Middle Ages, the *praguergroschen* of Bohemia arose in the early

fourteenth century to act as eastern Europe's silver multiple. It was widely used in 120
the region in combination with the gold ducats produced from the king of 121
Hungary's mine at Kremnica. Bohemian and German silver flowed eastwards as well as westwards, with Venice dominating the export of bullion to the Islamic world, though Venetian coinage also spread widely in the Balkans and was copied there in the thirteenth century, as the new kingdoms of Serbia and Bulgaria began to produce coinage of their own. The Venetian ducat brought gold to the eastern Mediterranean, inspiring imitations in the Latin states there and competing with Islamic *dinars* in Egypt and Syria. In the north-east the production of coinage began to penetrate Russia in the late fourteenth century, in addition to the silver ingots known as *grivnas* or *nubles* which still acted as the principal form of precious- 117
metal circulation there in the twelfth and thirteenth centuries.

117 Silver grivna of Russia, Novgorod type. Trading imbalances appear to have carried European silver in quantity to the Russian principalities via Novgorod, where it was recast into the ingots which functioned as the main means of exchange in Russia between the eleventh and fourteenth centuries. The local production and use of coin as money did not begin until the later fourteenth century.

The development of relatively stable international trading networks enabled systems to evolve which permitted the transfer of funds without the risk and inconvenience of moving huge amounts of coin. Money could be deposited, for instance, with the Order of the Temple or a Lombard merchant family, a credit note could be received and the equivalent sum in a specified form of currency (less charges) received at another place, where the same institution had an outlet. Large-scale trading systems could thus develop which minimised the transfer of physical sums of money. The fairs held in Champagne in northern France acted as a major international clearing house for the settlement of debts between Flanders and Italy. It must be stressed, however, that this network was relatively



118 Genoese bankers, from *De Septem Vitis*, a late fourteenth-century manuscript. Through the development of the bill of exchange the bankers of north Italy provided the international networks which permitted the transfer of funds across distances without requiring the movement of coin. Surviving Genoese registers record its origins in the late twelfth century and its full flowering in the thirteenth. Genoa was also a pioneer of advances in local banking, allowing transfers both between different accounts in the same bank and between different banks altogether. Genoese bankers also developed interest-bearing deposit accounts (British Library).

119 Gold florin of Humbert II, Dauphin of Vienne (1333–49). The minting of florin imitations began in Vienne in 1327, indicating the availability of Italian gold in the Rhone valley. This was encouraged by the arrival of the papal court in Avignon as the centre of a taxation system initiating bullion transfers across Christendom.

restricted in extent, with no more than a dozen or so cities forming its core. Lübeck and the Hanseatic German trading cities of the north were not part of it until the fifteenth century.

The great merchant houses of Tuscany and Lombardy made important contributions to the nature of medieval money, though this involved circumventing the entrenched theological prohibitions of 'usury' – making money breed by charging interest on a loan – which was regarded as being against divine law. Nicholas Oresme (d. 1382) in his *Treatise on the Origin, Nature, Law and Alterations of Money* is severe and clear in outlining the ideological restrictions underpinning the official medieval view of money: 'There are three ways in which profit may be made from money, without laying it out for its natural purpose: one is the art of the moneychanger, banking or exchange; another is usury; a third is alteration of the coinage. The first way is contemptible, the second bad, and the third worse.' Yet the Italians were able to develop deposit-banking and the bill of exchange, enabling banking transactions to be carried out at a distance. They also began to extend credit and lend on a much larger scale than their Jewish predecessors who had dominated money-lending in the early medieval period. Exchange contracts used in long-distance trade, where an element of risk permitted charges, could be used to disguise loans, for instance. Thus a bill of exchange could represent a simple transfer of funds, or else an investment transaction.

120 Silver praguergroschen of Wenceslas II, King of Bohemia (1278–1305). The praguergroschen became familiar across eastern Europe, and much Bohemian silver was exported east and south, instead of being coined. (× 2)

Necessity overcame many scruples, and popes, kings and other rulers were able to borrow against future income, giving them more leeway in the ever more expensive fields of warfare and diplomacy. This period also saw the development of public debt among the states of Italy, a crucial innovation for the history of banking and public finance. Between 1303 and 1400 the public debt of the city of Florence rose from 50,000 gold florins to 3 million. By allowing credit and overdrafts beyond the totals of their deposits, Italian bankers were able qualitatively to increase the money supply, releasing it from the age-old restrictions imposed by the amount of coinage currently in circulation. However, most of the great Italian banking or financial houses eventually overextended themselves, usually in royal loans they could hardly refuse, and crashed when their royal debtors defaulted. The kings of England and France between them caused the failure of the Riccardi of Lucca in 1294; Edward III of England ruined the Florentine houses of the Peruzzi in 1343 and the Bardi in 1346. The Medici were the greatest bankers of the later Middle Ages, but 118

121 Gold ducat of Ladislaus V, King of Hungary (1453–7), showing St Ladislas with orb and battle-axe. By 1328 enough gold was coming from Kremnica for a gold coinage to be introduced. Production was high throughout the later Middle Ages, dominating supplies in Central Europe, where Hungarian ducats were long the preferred means of payment. (× 2)

even they fell in 1494, when long-term trading imbalances tied up their funds away from Italy where they faced political pressures in Florence itself. Despite the refinements in banking and high finance, all imbalances ultimately still had to be settled in gold and silver.

122 Silver gigliato of Charles II, King of Naples (1285–1309). Larger grossi, double the size of the first Italian grossi, came to dominate Italian silver coin in the later Middle Ages. The mines of Sardinia provided the silver for the Neapolitan gigliato, which became an important Mediterranean trade coin, rivalling, and in the 1330s supplanting, the Venetian grosso.

Money was playing a much more important role in government and in society generally. Rulers increasingly paid their servants salaries rather than in grants of land. They ceased to rely on the old feudal levy to raise armies, which through tradition and precedent was too circumscribed and inflexible, and instead supplied and paid troops in cash. Larger-scale struggles, such as the Hundred Years' War between England and France (1337–1453), encouraged the subsidy of allies and payments of huge ransoms and dowries, while the Crusades and long-distance pilgrimages also necessitated the large-scale transmission of funds.

Feudal tenants increasingly commuted their labour services into cash rents, while their lords relied more on such cash payments or more directly and intensively exploited their estates to produce saleable surpluses. From the thirteenth century onwards networks of weekly markets made it possible for lesser monetary dealings to penetrate well outside the towns and cities. Prices in general rose occasionally, if nowhere near modern standards, but wages tended to keep pace, despite the interests of those benefiting from fixed rents and wedded to the concept of the just price and the just wage. The increased money supply was one

123 Gold pavillon of Philip VI, King of France (1328–50), showing the king beneath a canopy decorated with fleurs-de-lis. The introduction of gold coin to France was gradual. There were several issues, but only in the 1330s was there sufficient gold in the economy to ensure a satisfactory relationship between gold and silver. France's early gold issues had designs of high skill and set the style for other western rulers. (× 3)

124 Gold noble of Edward III, King of England (1327–77). The king is represented holding sword and shield and standing in a ship. By the 1340s foreign gold coins were familiar in England, and, with parliament's encouragement, the king introduced both local gold and silver multiple issues. The noble fitted into both of England's units of account, being half a mark and a third of a pound.

reason for such rises, though other factors contributed. For instance, labour shortages following the population collapse of the Black Death (1346–7) helped raise wages.

Yet we should not overestimate the use of coined money at this time. It was much more familiar to most people than in the early Middle Ages, but nevertheless much of society still would not have employed coin on a daily basis. Most rural daily life would have been based on a large degree of self-sufficiency and small-scale barter, with coin acquired occasionally on the sale of surplus agricultural produce and disbursed equally occasionally to pay taxes, rents, dues, fines or rare payments for specialist goods or services. These certainly grew more frequent and substantial, as cash rents took over, but the use of money might easily be virtually seasonal, following harvest time. This was reflected in the usual dates fixed for the payments of dues and rents, All Saints' Day (1 November) and the feast of St John the Baptist (24 June). Richer folk would have had a greater store of wealth and more constant access to funds, but this was still subject to the sheer availability of coin.

125 Gold saluto of Charles I, King of Sicily (1266–85), showing the Annunciation: the Archangel Gabriel approaches the Virgin Mary, a lily separating them. Religious imagery was common in medieval coin design. (× 2)

Liquidity could therefore be a major problem. As the fourteenth century advanced with no new silver deposits being discovered and the older ones becoming increasingly less productive, bullion supplies diminished as the metal circulated through the continent and beyond. The flow of silver to the Near East as coin or bullion was one of Europe's major exports in exchange for the spices (which made winter food palatable) and the other Eastern goods Westerners craved, and the movement of precious metal brought great profit to middlemen like the Venetians as well as to the silver-producers themselves. Local shortages were always possible, if local mint prices were uncompetitive, or if local produce was in little demand. Bimetallism, the use of gold and silver coinage concurrently in an integrated currency system, could enhance such problems: some rulers tried to fix the gold/silver ratio at an unrealistic level, causing undervalued metal to be taken to a mint that offered a better price. But as the fifteenth century arrived, there were longer periods of severe money shortage, the so-called 'bullion famine', which curtailed trade and other economic activity, even for those rich in property and other goods. Thus, a system accustomed to, and often requiring, the availability of coin was faced with severe shortages of hard cash.

126 Gold gulden of Adolf II von Nassau, Archbishop of Mainz (1461–75), showing the arms of the Rhineland electors, the archbishops of Mainz, Cologne and Trier, and the Count Palatine. In 1385 these princes founded the Rhenish Monetary Union, co-ordinating their coinages and often their designs. This design represents the eighteenth Treaty of the Union, introduced in 1464 to last for twenty years.

While mints of different rulers competed for available bullion, there were also pressures to link different currencies to facilitate ongoing commerce. In the fourteenth century the coinages of the Rhineland were linked by a series of 126

127 Silver witten of Lübeck, *c.*1366–80. Monetary conventions linked the towns of the Hanseatic League, beginning with Lübeck and Wismar, followed by Hamburg, Lüneburg and others. The first *Rezess* (convention) of the union was held in 1379, and successive such agreements regulated north German currency in the name of the Wendish Monetary Union until 1569. (× 2)

conventions between the region's princes, who agreed to strike their gold coinage, and sometimes part of their silver coinage, to a common standard. Similarly, from the mid-fourteenth century several of the cities of the Hanseatic League on the Baltic coast issued a silver coin (the *witten*) to common standards 127 in the Wendish Monetary Union. In fifteenth-century Italy many smaller coin issuers produced ducats to the Venetian standard. Political developments added to this trend. The House of Burgundy obtained control of most of the secular 128 principalities of the Netherlands (Holland, Flanders, Hainaut, Brabant and Luxemburg) and introduced a common coinage for the region; the various kingdoms of the Iberian peninsula (except Portugal) fell to a single ruling house, and Castilian coinage became the national coinage of Spain in the 1490s; and a network of royal mints across France reinforced the ubiquity of the national 130 currency, restored to stability after the tribulations of the Hundred Years' War.

By the mid-fifteenth century the tight monetary squeeze caused by the bullion famine was easing, a consequence of new discoveries of silver and developments in mining techniques. The impact of further new bullion supplies from the Americas was to transform money in Europe after the Middle Ages.

129 Gold ducato di camera of Pope Innocent VIII (1484–92), showing St Peter fishing. Papal coinage only became extensive in the later Middle Ages, following the move to Avignon. It expanded in Italy in the later fifteenth century, when the Papal States became more ordered. Two extensive series of gold ducats were struck, one based on the Venetian coin and the other (the ducato di camera) slightly lighter. (× 1½)

128 Gold lion of Philip the Good, Duke of Burgundy, as Count of Flanders (1419–67), named after its design of a lion beneath a canopy. The main principalities of the Netherlands fell into the hands of the dukes of Burgundy, and a unified coinage (covering Flanders, Brabant, Hainaut and Holland) was first issued in 1434; the lion became the main gold coin in 1454 under a revision of the system. (× 2)

130 Gold écu au soleil of Louis XI, King of France (1461–83). The name of the denomination was derived from its design: the crowned shield (*écu*) of France under the sun (*soleil*). The écu was the principal gold coin of France from 1385 well into the early modern period, subject to relatively minor alterations. Secret marks indicate which of the various French mints struck each coin (here an annulet under the *v* of *Ludovicus* indicates Toulouse). (× 2)